

5. A set of attrib-ids

An attribute on a given server cluster instance is associated with an Atomic Write State if the state's set of attrib-ids contains the ID of the attribute, the cluster-id matches the ID of the cluster instance and its endpoint-no matches the endpoint of cluster instance.

A client is associated with an Atomic Write State if both its [Atomic Writer ID](#) and [accessing fabric](#) match the [Atomic Writer ID](#) and [accessing fabric](#) of the client.

If a server receives a Write Request for an attribute that is not associated with an Atomic Write State that is also associated with the client making the request, the server SHALL return the error code INVALID_IN_STATE.

Reading an attribute with the Atomic quality from a client with an associated Atomic Write State SHALL return the current value of the attribute if the client has not successfully written to the attribute; if the client has written to the attribute while having an associated Atomic Write State, the server SHALL return the updated value of the attribute.

When writing to any attribute associated with an Atomic Write State, any integrity checks SHALL be evaluated in the context of the pending values of all the attributes associated with the Atomic Write State.

Any writes to attributes with the Atomic quality from a client associated with an Atomic Write State SHALL only be visible to that client until committed, and SHALL NOT have any effect on the operation of the server.

If a server receives a command which modifies an attribute with the Atomic quality while a client has an Atomic Write State containing the attrib-id of that attribute, it SHALL apply these changes as if there were no atomic write; this MAY cause the atomic write to fail when the client attempts to commit (see [Commit Write](#)).

7.15.4. AtomicRequestTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	BeginWrite	Begin an atomic write	M
1	CommitWrite	Commit an atomic write	M
2	RollbackWrite	Rollback an atomic write, discarding any pending changes	M

7.15.5. AtomicAttributeStatusStruct Type

This struct indicates the status of an attribute during an atomic write.